

Arthroscopic Free Bone Block Transfer Technique Using Iliac Crest Autograft With Anchor-Button Fixation for Glenohumeral Instability and Glenoid Bone Loss

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Abstract: Bone block transfer for the treatment of glenohumeral instability with severe glenoid bone loss or multiple recurrences has grown in popularity in recent years. In this technical note, we described an arthroscopic-assisted free bone block transfer technique, using a specially designed anchor-button unit to treat glenohumeral instability caused by Bankart lesions accompanying glenoid bone loss. This approach is intended to accomplish mechanical restoration while minimizing subluxation and recurrence, especially in the critical early postoperative phase, through an anchor-button locking mechanism that offers the most significant advantage of this technique: more rigid fixation and improved rotational stability of the autologous iliac crest graft compared with conventional techniques relying on screws and suture anchors.

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Glenohumeral instability, especially with a substantial glenoid bone defect, presents a considerable challenge in shoulder surgery because simple Bankart repair results in high recurrence rates.¹ Achieving long-term stability requires bony reconstruction operations like Latarjet and free bone block transfer.² The threshold for glenoid bone loss necessitating augmentation has long been debated, with current literature suggesting values as low as 13.5% in high-demand patients.³ As this threshold decreases, the number of bone augmentation procedures is expected to rise.

Free bone block transfer is employed for reestablishing preinjury sports performance with minimal postoperative movement restriction while preserving the coracoid process and subscapularis.⁴ Previous techniques, which rely on screws and suture anchors, have shown stability in various studies.⁵ Despite technical advancements, the graft's rotational stability, particularly in the early postoperative period, remains a concern.⁶

This technical note describes an arthroscopic-assisted bone block transfer technique that uses an anchor-button design (Doratek Medical, Ankara, Turkey) for the

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Arthroscopic Free Bone Block Transfer Technique using Iliac Crest Autograft with Anchor-Button Fixation for Glenohumeral Instability and Glenoid Bone Loss

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VIDEO 1 Labrum and the glenoid cavity are examined arthroscopically by the senior surgeon. The labrum remnants are mobilized, and the bone defect on the anterior glenoid is prepared for implantation by removing soft tissue and smoothing the surface. Meanwhile, the second surgeon begins harvesting a tricortical iliac bone graft with dimensions of 2×1 cm. The graft's shape is adjusted to form a regular rectangular prism measuring $16 \times 8 \times 6$ mm. Two tunnels are formed in the bone block at appropriate intervals to correlate with the holes on the button, and the button is placed over the holes on the cortical side of the graft. In the meantime, the rotator interval is excised to allow the bone block to pass through. Then, 2 specially designed anchors with backward protrusions are inserted into the bone loss site using a special guide. A new anterior portal is created, and a large cannula is placed into it, passing through the rotator interval. A transport system is established by using the sutures from previously inserted anchors and tunnels in the graft. Using this system, the bone block with the button on it is placed on the anchor protrusions at the defect site. The button's holes fit and lock over the anchor protrusions that pass through the tunnels of the graft. Sutures from the transporter system are employed to further secure the graft. Finally, a Bankart repair is performed on the glenoid over the bone block. Video content can be viewed at <https://doi.org/10.1002/atn2.70250>.

management of glenohumeral instability with bone loss. We believe that this approach promises to provide anatomical restoration with robust fixation and enhanced rotational stability compared with conventional screw or suture anchor techniques.

SURGICAL TECHNIQUE

This study does not require any institutional board review or approval. This study is not a randomized controlled trial.

Patient Positioning

The patient, under general anesthesia, is positioned laterally on the contralateral side, with a pillow placed

between the knees. Anterior and posterior curved supports sustain the patient in this position. Traction equivalent to approximately 5% of the patient's body weight is applied to the arm at 45° abduction and 15° flexion. The surgical draping is applied after the preparation of the skin on the shoulder and the ipsilateral iliac crest regions as seen in Figure 1.

Shoulder Arthroscopy and Preparation for Graft Implantation

Routine arthroscopic examination is started with viewing through the posterior portal. The labrum remnant on the anterior side is mobilized using a rasp, and the glenoid defect surface is prepared both biologically and mechanically. Fibrous or sclerotic tissue on the surface, as observed in chronic lesions, should be excised using a shaver or burr (Table 1). Then, the area is leveled to achieve a planar surface across the length and width of the intended bone graft (Figure 2A).

The rotator interval is excised through another portal anteriorly to facilitate the passage of the bone block. A special guide (Doratek Medical, Ankara, Turkey) featuring 2 tunnels for placing suture anchors at the same interval with the holes in the plate is inserted through the rotator interval. The guide also enables the surgeon to assess the bone block level relative to the glenoid cartilage surface. Two suture anchors with backward-facing titanium protrusions are implanted into the prepared defect surface (Figure 2B).

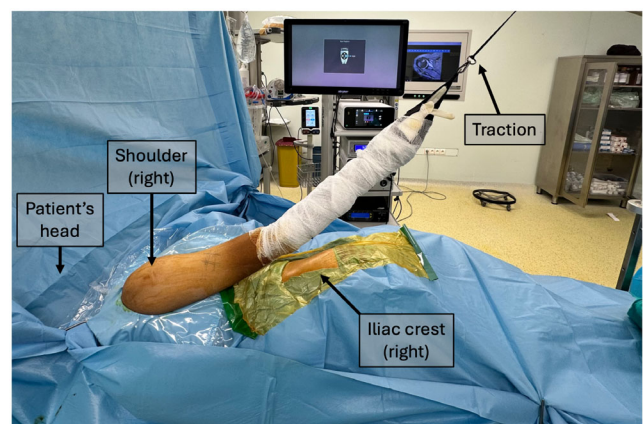


FIGURE 1 The patient is in the lateral decubitus position, with traction equivalent to about 5% of the patient's body weight administered to the right arm at 45° abduction and 15° flexion, as seen from the surgeon's position. The patient is supported in this position by anterior and posterior curved supports on the chest, interscapular area, and lower back, as well as a pillow between the flexed knees. The surgical drape is applied after the skin on the shoulder and ipsilateral iliac crest has been prepared.

TABLE 1 Pearls and Pitfalls of the Arthroscopic Free Bone Block Transfer Technique Using Anchor-Button Fixation**Pearls**

- Prepare the glenoid defect surface thoroughly by removing all fibrous and sclerotic tissue to achieve a planar surface for optimal graft contact
- Place the anchors at the same interval as the button holes, ensuring accurate alignment before graft insertion
- Ensure the spongy surface of the graft faces the spongy surface of the defect
- Allocate sufficient time to position the graft-button complex accurately on the anchor protrusions before securing with a nonsliding knot
- Verify that no labral or soft tissue remains between the glenoid and the graft before final fixation
- Apply pressure to the button with a blunt instrument while tying the knot to ensure compression of the graft against the glenoid

Pitfalls

- Failure to adequately prepare the defect surface may result in poor graft-bone contact and compromised healing
- An oversized or irregularly shaped graft will not pass through the 12-mm cannula or fit the defect properly
- Misalignment between anchor protrusions and button holes will prevent proper locking and compromise stability
- Insufficient compression during knot tying may result in graft loosening or micromotion in the early postoperative period

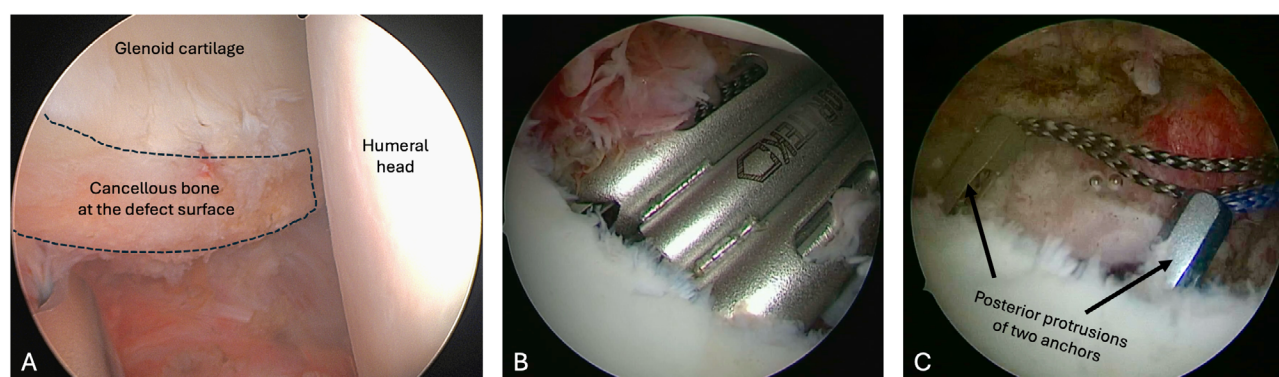


FIGURE 2 (A) Protrusions on the surface were removed with a burr to provide a level surface across the length and width of the proposed bone transplant. This figure shows the prepared glenoid defect surface viewing through the anterior portal. (B) Viewing through the anterior portal, a specific guide (Doratek Medical, Ankara, Turkey) is used to place 2 anchors separated by a certain distance. (C) Two anchors with backward-facing protrusions are inserted at the same distance as the button's holes.

Harvesting and Preparation of Autologous Iliac Crest Graft

A second surgeon may begin harvesting the bone graft concurrently. A skin incision of about 5 cm begins 2 cm posterior to the anterior superior iliac spine and extends posteriorly over the iliac crest. After dissection through the skin and subcutaneous tissue, a tricortical bone graft of approximately $2 \times 1 \times 1$ cm is harvested without the periosteum. The harvested graft is subsequently cut to the final dimensions of $16 \times 8 \times 6$ mm using a fine-tipped saw (Figure 3A). The button (Doratek Medical, Ankara, Turkey) is positioned on the cortical surface with 16×8 mm dimensions, and 2 holes are drilled in the graft

to align with the plate's holes, each measuring 2.7 mm in diameter (Figure 3B).

The rotator interval is excised, and a large cannula (Doratek Medical, Ankara, Turkey) with a diameter of 12 mm which allows the passage of the bone block is inserted by enlarging the existing portal or a new portal is established (Figure 4). Alternatively, a cannula can be created by cutting the distal tip of a 10-cc syringe.

Establishment of the Transporter System for Bone Block

One free suture from each anchor on the glenoid is passed ex situ through the holes on the bone block-button

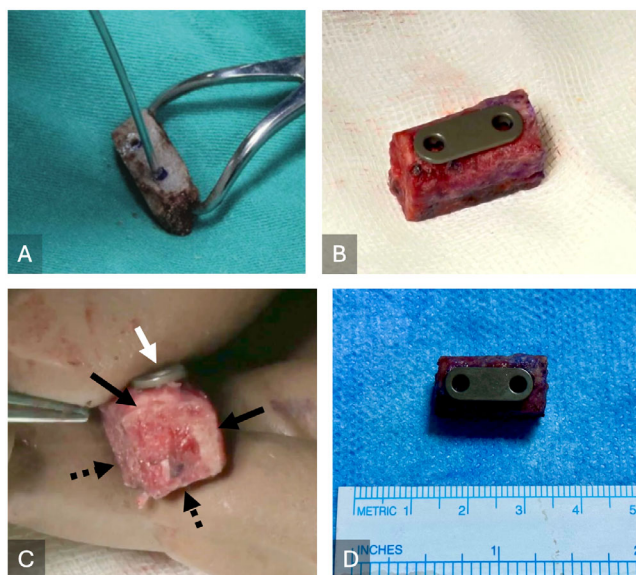


FIGURE 3 (A) The tricortical iliac bone graft harvested from the patient is trimmed to the final dimensions of $16 \times 8 \times 6$ mm with a fine-blade saw. (B) The special button is placed on the cortical surface, and two 2.7-mm-diameter tunnels are drilled in the graft to align with the plate holes. (C) A transverse view from the graft, showing 2 cortical (black arrows) and 2 spongy (dotted arrow) osseous surfaces. The button (white arrow) is placed on the drilled cortical surface. The other ends of the tunnels are on the graft's opposite side cancellous surface. (D) A top view of the button-graft unit with a ruler.

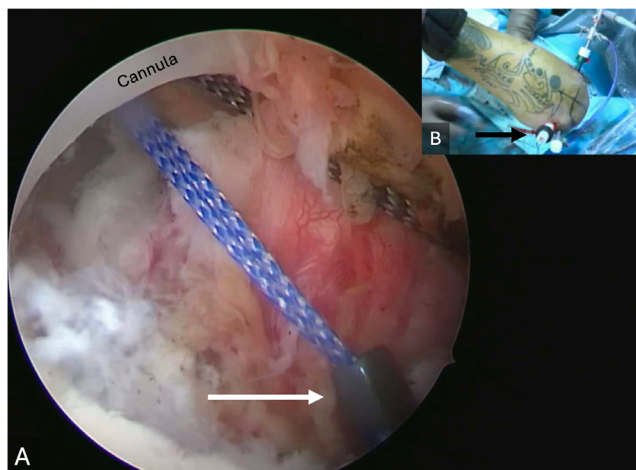


FIGURE 4 Viewing through the posterior portal, the rotator interval is excised, and a large cannula with a diameter of 12 mm which allows the passage of the bone block is placed into a new portal here. (A) Arthroscopic view of the large cannula after sutures of the anchors were transported into it. The white arrow points to the backward protrusion of the suture anchor. (B) Anterior side view with the external camera, showing the large white cannula (black arrow) with a black ring for greater control and maneuverability, inserted through an anterior portal to the right shoulder.

complex (Video 1). The orientation of the bone block must be adjusted to ensure that the cancellous surface of the graft contacts with the cancellous surface of the defect area, whereas the cortical surface of the graft, the one without the button, is positioned parallel to the articular cartilage. The suture from the superior anchor is threaded through the superior hole of the bone block, whereas the suture from the inferior anchor is threaded through the inferior. These sutures are tied in a figure-of-eight knot over the button, and the slack is removed by pulling their free ends (Figure 5).

Transportation and Implantation

The graft is placed in the cannula with its superior tip facing forward and moved into the joint by alternately pulling the free suture ends of the carrier system. The graft-button complex is gently manipulated with a probe and lined up on the anchors. It is critical to ensure that no labral or any other soft tissue remains between the glenoid and the graft. The surgeon must allocate time to position the graft-button complex accurately, ensuring alignment of the holes with the anchor protrusions for proper locking (Figure 6A). After confirming the alignment of the graft's cortical surface with glenoid cartilage, a nonsliding knot is secured using the free ends of the sutures of the carrier system, whereas pressure is applied to the button with a blunt instrument (Figure 6B). Finally, the labral remnant is fixated on the glenoid over the bone graft using the remaining sutures on anchors or using additional conventional suture anchors (Figure 6C).

Postoperative Rehabilitation

Patients are advised to wear a neutral shoulder sling for the first 6 weeks. Shoulder passive range-of-motion exercises commence post the initial week. During the early postoperative period, internal rotation with horizontal adduction and external rotation with abduction (apprehension position) are strictly avoided. Active shoulder range-of-motion exercises begin after the sixth week, focusing on the gradual restoration of normal external rotation range over time. Strengthening exercises with resistance are initiated starting in the third month postoperatively. Patients may return to unrestricted daily activities and begin progressive, sport-specific exercises after 6 months.

DISCUSSION

The presented surgical technique describes arthroscopic bone block transfer with an anchor-button complex

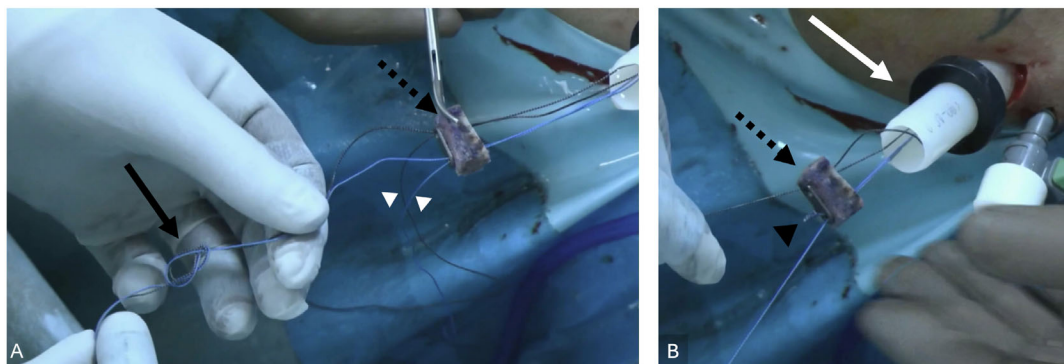


FIGURE 5 Anterior side view with the external camera. (A) One end of the sutures from each anchor is tied with an 8-shaped knot (black arrow) to be placed over the graft-button unit (dashed arrow). Other ends (white arrowheads) of these sutures remain free. (B) The eight-shaped knot is secured, and increased parts of the sutures were removed. The loose parts of the sutures are tightened by pulling their free ends, and the knot (black arrowhead) is placed over the graft-button unit (dashed arrow). Then, the unit is placed inside the large cannula (white arrow) with its end with the inferior anchor's sutures placed forward.

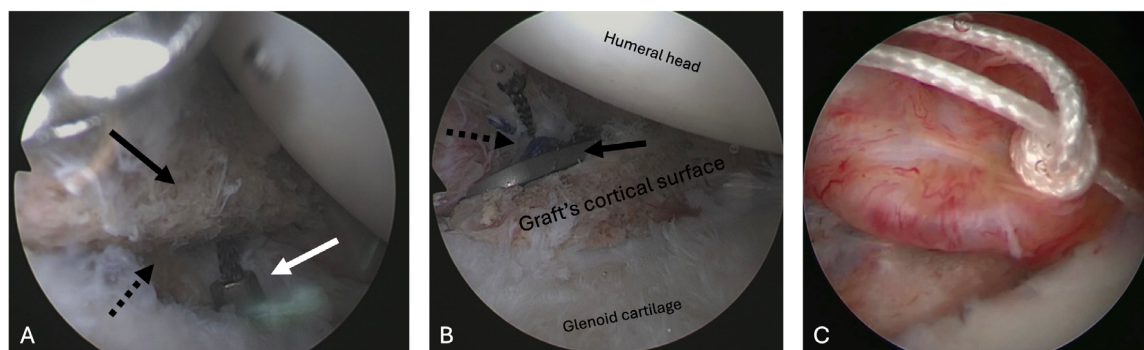


FIGURE 6 Views through the posterior portal. (A) The surgeon must allocate time to position the graft-button complex accurately, ensuring alignment of the holes with the anchor protrusions (white arrow) for proper locking. The cancellous surface of the graft (dotted arrow) must face the glenoid's prepared cancellous surface, whereas the cortical surface (black arrow) faces the joint cavity. (B) After confirming the alignment of the graft's cortical surface with glenoid cartilage, a nonsliding knot is secured (dotted arrow) using the free ends of the sutures of the carrier system, whereas pressure is applied to the button (black arrow) with a blunt instrument. (C) Finally, the labral remnant or the capsule is fixated on the glenoid over the bone graft using the remaining sutures on each anchor or new suture anchors.

for glenohumeral instability. This technique is distinguished by the use of a specially designed button that locks with the anchors for fixation.

Recent literature indicates that the threshold for “critical” bone loss relative to the native glenoid surface has decreased from 25% to about 15% and even to 13.5% in high-demand athletes.³ Consequently, the frequency of bony procedures is expected to increase in the imminent future.

There are various surgical procedures recommended for the treatment of patients with significant bone loss or recurring instability after a Bankart repair, such as in patients with epilepsy. Arthroscopic Latarjet procedure and arthroscopic iliac crest bone block transfer are the 2 commonly used methods, presenting distinct advantages and disadvantages.² Arthroscopic Latarjet procedure is

suggested even in the first shoulder dislocation cases by some authors.^{7,8} The arthroscopic technique was designed to prevent complete incision of the subscapularis and complications related to the brachial plexus in open Latarjet surgery; however, the steep learning curve and the challenges in accurately identifying the correct level for splitting the subscapularis complicate the procedure.

Free iliac bone block transfer is primarily used for large bone defects unsuitable for coracoid grafts and for revisions after the Latarjet procedure.⁹ Arthroscopic free bone block transfer technique was developed to mitigate neurovascular complications of open surgery¹⁰ (Table 2). A randomized controlled trial comparing the Latarjet procedure to the arthroscopic free bone block transfer technique found no statistically significant difference in outcomes.¹¹ Research has also revealed that the Latarjet procedure is

TABLE 2 Advantages and Disadvantages of the Arthroscopic Free Bone Block Transfer Technique Using Anchor-Button Fixation

Advantages	
Anchor-button locking mechanism provides more rigid fixation compared with conventional techniques	
Enhanced rotational stability of the bone graft, particularly in the early postoperative period	
Autologous iliac crest graft preserves the coracoid process and subscapularis integrity	
Arthroscopic-assisted approach reduces neurovascular complications compared with open surgery	
Facilitates return to preinjury sports performance with minimal external rotation restriction	
Disadvantages	
The technique involves a learning curve for establishing the transporter system and aligning the graft-button complex on the anchor protrusions	
Requires specially designed anchors and button (Doratek Medical), which may not be widely available	
Donor-site morbidity from iliac crest harvest, including potential sensory disturbance	
Requires a second surgeon or sequential approach for graft harvesting	
Absence of long-term clinical outcome data	
Metallic hardware-related complication risk	

associated with a higher incidence of internal rotation limitation, whereas free iliac bone block transfer is linked to sensory problems in the donor area. Arthroscopic free bone block transfer preserves the integrity of the coracoid process and subscapularis, facilitating return to preinjury athletic performance in most competitive athletes with minimal external rotation restriction.⁴

The literature indicates a higher recurrence of instability after revision with free bone block procedures relative to the Latarjet procedure; however, the incidence of a second revision is lower with bone block procedures.¹¹ This controversy may stem from the definition of recurrent instability, which often includes apprehension and subluxations alongside frank dislocations. Despite the absence of overt dislocation, subluxation and associated complaints can be attributed to insufficient rotational stability of the graft and minimal rotation, particularly in the early postoperative period. Numerous biomechanical studies have assessed the stability of bone grafts for glenoid defect reconstruction, showing that sutures provide adequate stability. Several biomechanical studies have assessed bone graft stability for glenoid defect reconstruction;¹²⁻¹⁴ however, none have specifically examined early rotational stability.

In the context of free bone block application, particularly when comparing our anchor-locking block to previous suture anchor techniques, the enhanced stability promised with our approach is noteworthy. The design rationale for the current implant and technique application aims to attain enhanced rigid fixation and rotational stability while minimizing intra-articular bulk. The design

offers enhanced rotational rigidity in fixation, crucial for minimizing subluxations and recurring instability, particularly during the early postoperative phase. This technique is designed for the most common type of instability, anterior.¹⁵ Yet, it offers a general framework applicable to instability in other directions.

In conclusion, this technique with an anchor-button unit attempts to reduce recurring instability and improve long-term outcomes for patients with glenohumeral instability accompanying glenoid bone loss. The design of the anchor-locking mechanism offers enhanced rotational stability, making it a viable alternative that addresses a fundamental weakness of previous free bone block procedures.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the author(s) used the Eleven v3 (Expressive) model (ElevenLabs, New York, NY) to generate audio narration for the surgical technique video. The script was completely authored and verified by the surgical team before voice synthesis.

DISCLOSURES

The authors (B.B., B.S., M.F.T., I.H.A., H.Y., M.O., U.K.) declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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